

Dimitrios Panagiotis Emmanouilidis

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Software developer — AI engineering: local LLM inference, retrieval, evaluation.

TECHNICAL SKILLS

Languages: Python, C++ , Kotlin

Platforms & Tooling: Android (NDK, JNI, Gradle), Linux, Windows, Git, GPU/VRAM budgeting, cross-platform toolchain debugging

AI & Machine Learning: Ollama, llama.cpp/GGUF, local LLM inference, quantised models, hybrid retrieval (dense + lexical), JSON Schema validation, evaluation harness design

TECHNICAL PROJECTS

actuation-runner — LLM Action-Payload Failure Benchmark

Sole Author · Public · Aug 2026 · github.com/dpemmanouilidis/actuation-runner

- Open-source benchmark measuring how often LLM-generated action payloads fail before reaching an actuator, classified across eight outcome categories (one pass, seven distinct failure modes).
- Two independent 100-trial runs against a local quantised 9B model: 89/100 and 92/100 valid. Every failure across all 200 trials was planner token-ceiling truncation — a one-to-one correspondence with the model's length stop reason, with no exceptions.
- Run-to-run spread of 11 and 8 failures on denominators of 100 is itself the finding: either run alone would have read as a precise figure.
- 45-test offline suite, including tests that invoke the CLI as a subprocess rather than importing it, so the real invocation path is exercised end to end.
- Three defects caught before the measured run by reading source and showing computed inputs prior to execution — among them a CLI entry point that printed results while writing nothing to disk, with 23 tests passing against it at the time.
- Python, jsonschema, Ollama. Apache 2.0.

Offline Local Inference & RAG Application

Lead Developer · In Development · Private repo, walkthrough available on request

- Android app running a quantised LLM entirely on-device, with no network at query time; llama.cpp/GGUF core over FFI behind a native Jetpack Compose UI.
- Hybrid retrieval fusing 384-dim MiniLM dense vectors with a character-trigram BM25 index. Answers carry passage citations and refuse when the corpus does not contain the answer.
- Footprint budgeted against Google Play asset delivery: a 46 MB embedding model beside a 2B-class Q4_K_M LLM at about 1.3 GB, inside the 1.5 GB per-pack limit. Verified on a Samsung Galaxy S24.
- Built the retrieval eval harness that gates release, including a hand-curated catastrophic-miss list that blocks a corpus build from shipping regardless of average recall.
- Kotlin, C++ , llama.cpp, all-MiniLM-L6-v2, Room/SQLite, Python build pipeline.

Contextual Synthesis and Constrained Action: Evaluating a Two-Agent LLM System for Personalized Vehicle Adaptation

BSc Dissertation · Jan–May 2026

- Two-stage LLM pipeline (profiler feeding planner) behind a non-LLM JSON Schema safety gate, tested in the CARLA simulator.
- 4-bit quantised 9B model via Ollama in a 12 GB VRAM budget shared with the simulator.
- Measured end-to-end latency and reported it as prohibitive for real-time control on that hardware.
- Found an idle simulator process halved local inference throughput — roughly 51 tok/s against 93 tok/s — with GPU utilisation still reporting 100%.

EDUCATION

BSc (Hons) Computing (Application Development)

New York College / University of Greater Manchester — Thessaloniki, Greece

Graduated July 2026